

Chinook SQL Analysis Project Documentation

Project Overview

This project focused on analyzing the Chinook music store database using PostgreSQL. The aim was to strengthen practical SQL skills by working with a relational database and answering business-related questions using different SQL techniques. Throughout the project, I applied joins, subqueries, window functions, aggregate functions, and query optimization concepts to extract meaningful information from the database..

Database Setup

The Chinook database was imported into PostgreSQL using pgAdmin.

During the import process, the original SQL script contained PostgreSQL command-line statements that were not supported by the pgAdmin Query Tool. These statements were removed before executing the script successfully.

After the database was imported, the tables were verified to ensure that all required data had been loaded correctly before beginning the analysis.

Advanced SQL Concepts

Joins

Different join operations were used to combine data from multiple related tables.

The following joins were practiced during the project:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN

These joins made it possible to retrieve customer information, invoice records, and track sales from different tables while maintaining the relationships between them.

Subqueries

Subqueries were used to answer questions that required intermediate calculations before producing the final result.

Examples included:

- Identifying customers whose spending was above the average customer spending.
- Finding customers who made more purchases than the average customer.

Using subqueries helped simplify complex analytical questions into smaller and more manageable steps.

Window Functions

Window functions were implemented using:

- ROW_NUMBER
- RANK
- PARTITION BY
- ORDER BY

These functions were used to assign sequential invoice numbers for each customer while keeping every invoice record in the result set.

This demonstrated how window functions can perform calculations across groups of rows without reducing the dataset.

Business Problem Solving

1. Top 5 Customers by Revenue

Purpose

The objective of this query was to identify the customers who generated the highest revenue for the business.

SQL Techniques Used

- INNER JOIN
- GROUP BY
- SUM
- ORDER BY
- LIMIT

Insight

The analysis showed that a small number of customers contributed the highest total revenue. These customers represent valuable business relationships and could be targeted for customer retention and loyalty programs.

2. Top 5 Tracks by Revenue

Purpose

This query was used to determine which music tracks generated the highest revenue based on customer purchases.

Revenue was calculated using:

$$\text{Revenue} = \text{Unit Price} \times \text{Quantity Sold}$$

SQL Techniques Used

- INNER JOIN
- Aggregate Functions
- GROUP BY
- SUM
- COUNT
- ORDER BY
- LIMIT

Insight

The results identified the highest-performing tracks in the database. These tracks generated the greatest revenue and represent the most commercially successful products.

3. Revenue Trends Over Time

Purpose

The purpose of this analysis was to examine how revenue changed over time using invoice dates.

The EXTRACT function was used to group revenue by year and month.

SQL Techniques Used

- EXTRACT
- SUM
- GROUP BY
- ORDER BY

Insight

Revenue remained relatively stable throughout the years covered by the dataset. Although there were small fluctuations between different periods, there was no significant decline or rapid growth, indicating consistent sales performance.

4. Customer Purchasing Behaviour

Purpose

This query analyzed customer purchasing behaviour by calculating:

- Number of purchases made by each customer.
- Total amount spent by each customer.

SQL Techniques Used

- INNER JOIN
- COUNT
- SUM
- GROUP BY
- ORDER BY

Insight

The results showed that several customers made the same number of purchases but spent different amounts overall. This indicates that purchase frequency alone does not determine customer value, as some customers spend significantly more per transaction than others.

Query Optimization

Basic query optimization concepts were demonstrated by creating indexes on frequently used columns.

Indexes were created on:

- `invoice(customer_id)`
- `invoice(invoice_date)`

These indexes help improve query performance by allowing PostgreSQL to locate matching records more efficiently during joins and date-based analysis.

In addition to indexing, all SQL queries were written using consistent formatting, meaningful aliases, descriptive column names, and comments to improve readability and maintainability.

Skills Demonstrated

Throughout this project, the following SQL skills were demonstrated:

- PostgreSQL
- Data Retrieval
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- Aggregate Functions
- GROUP BY
- HAVING
- Subqueries
- Window Functions
- Business Data Analysis
- Revenue Analysis
- Customer Behaviour Analysis
- Query Optimization

Conclusion

This project provided practical experience in analyzing a relational database using PostgreSQL. By applying advanced SQL techniques, I was able to answer business questions, identify customer and product performance, analyze revenue trends, and explore customer purchasing behaviour.

The project also reinforced the importance of writing clean, organized, and efficient SQL queries while applying basic optimization techniques to improve performance.